Homework 05: Sample solution

QMIR

October 7, 2026

Table of contents

1 Explore 3

2 The grammar of graphics 5

3 One variable 9

4 Two variables 15

5 Three and more variables 19

6 Styling: less ink, more information 22

7 Common pitfalls: what is wrong here? 26

8 Your figure: a data summary 31

Reflection 34

Session info and execution time 34


```
start_time <- Sys.time()

# Install once, in the Console, not here:
# install.packages("pak")
# pak::pak("vdeminstitute/vdemdata")

library(tidyverse) # wrangling and visualization
library(janitor) # quick tables with tabyl()
library(vdemdata) # the V-Dem dataset
```

```r
set.seed(2026)

# One row per country and year, from 1950 on.
vdem_df <- vdem |>
  as_tibble() |>
  filter(year >= 1950) |>
  transmute(
    country = country_name,
    year,
    continent = case_match(
      e_regiongeo,
      1:4 ~ "Europe",
      5:9 ~ "Africa",
      10:14 ~ "Asia",
      15 ~ "Oceania",
      16:19 ~ "Americas"
    ),
    regime = factor(
      case_match(
        v2x_regime,
        0 ~ "Closed autocracy",
        1 ~ "Electoral autocracy",
        2 ~ "Electoral democracy",
        3 ~ "Liberal democracy"
      ),
      levels = c(
        "Closed autocracy",
        "Electoral autocracy",
        "Electoral democracy",
        "Liberal democracy"
      )
    ),
    libdem = v2x_libdem,
    polyarchy = v2x_polyarchy,
    liberal = v2x_liberal,
    egaldem = v2x_egaldem,
    partipdem = v2x_partipdem,
    freexp = v2x_freexp_altinf,
    corruption = v2x_corr,
    women = v2x_gender,
    gdp_pc = e_gdppc,
    pop = e_pop / 100
```

```
  )
```

1 Explore

1. `glimpse()` `vdem_df`. How many rows and columns does it have, and what is one row?
2. How many countries does it cover, and which years? `count(country)` returns one row per country, as in homework 04, so `nrow()` of it is the number of countries. `summarise()` with `min()` and `max()` returns the first and the last year.

```
glimpse(vdem_df)
```

```
Rows: 12,646
Columns: 14
$ country    <chr> "Mexico", "Mexico", "Mexico", "Mexico", "Mexico", "Mexico",~
$ year       <dbl> 1950, 1951, 1952, 1953, 1954, 1955, 1956, 1957, 1958, 1959,~
$ continent  <chr> "Americas", "Americas", "Americas", "Americas", "Americas",~
$ regime     <fct> Electoral autocracy, Electoral autocracy, Electoral autocra~
$ libdem     <dbl> 0.095, 0.095, 0.095, 0.095, 0.104, 0.105, 0.105, 0.105, 0.1~
$ polyarchy  <dbl> 0.196, 0.196, 0.194, 0.195, 0.229, 0.230, 0.231, 0.231, 0.2~
$ liberal    <dbl> 0.263, 0.263, 0.263, 0.263, 0.267, 0.267, 0.267, 0.267, 0.2~
$ egaldem    <dbl> 0.098, 0.098, 0.097, 0.097, 0.106, 0.106, 0.106, 0.106, 0.1~
$ partipdem  <dbl> 0.122, 0.122, 0.123, 0.123, 0.133, 0.133, 0.133, 0.133, 0.1~
$ freexp     <dbl> 0.381, 0.381, 0.362, 0.362, 0.362, 0.362, 0.362, 0.362, 0.3~
$ corruption <dbl> 0.82, 0.82, 0.82, 0.82, 0.82, 0.82, 0.82, 0.82, 0.82, 0.82,~
$ women      <dbl> 0.253, 0.253, 0.277, 0.277, 0.303, 0.314, 0.314, 0.319, 0.3~
$ gdp_pc     <dbl> 7.521, 7.803, 7.982, 8.058, 8.329, 8.648, 8.974, 9.277, 9.6~
$ pop        <dbl> 29.69409, 30.74441, 31.74480, 32.71254, 33.67513, 34.65725,~
```

```
vdem_df |>
  count(country) |>
  nrow()
```

```
[1] 182
```

```
vdem_df |>
  summarise(first_year = min(year), last_year = max(year))
```

```
# A tibble: 1 x 2
  first_year last_year
       <dbl>     <dbl>
1       1950      2025
```

`vdem_df` has 12646 rows and 14 columns, and one row is **one country in one year**. It covers 182 countries from 1950 to 2025. `count(country)` also shows how many years each country is observed, which is a quick check that the panel is not balanced. Not every country is observed in every year: states appear when V-Dem starts coding them and disappear when they dissolve, such as the German Democratic Republic in 1990.

3. Answer in prose: data that observe the same units again and again over time are called **panel data**. Name one question you could answer with `vdem_df` that you could not answer with data from a single year.

Any question about change within a country needs the panel. For example: did the countries that became electoral democracies in the 1990s stay democratic, or did some of them slide back? A single year shows where countries stand, but not where they came from.

4. Create `vdem_2025`, the rows of the most recent year, and make a quick table of `regime` with `tabyl()` and `adorn_totals()`, as in class. Which regime type is the most common in 2025?

```
vdem_2025 <- vdem_df |>
  filter(year == max(year))

vdem_2025 |>
  tabyl(regime) |>
  adorn_totals()
```

```
              regime   n   percent
    Closed autocracy  35 0.1955307
 Electoral autocracy  57 0.3184358
 Electoral democracy  56 0.3128492
   Liberal democracy  31 0.1731844
               Total 179 1.0000000
```

Electoral autocracy is the most common regime type in 2025, with 57 of 179 countries, just ahead of electoral democracy with 56. The most common regime in the world today holds elections that are not free and fair.

5. `gdp_pc` and `pop` end in 2019. Check this: count the non-missing values of `gdp_pc` in every year from 2016 to 2022. Hint: `summarise(n_gdp = sum(!is.na(gdp_pc)))` after a `group_by(year)`.
6. Create `vdem_2019`, the rows of 2019. Every figure that uses `gdp_pc` or `pop` uses `vdem_2019`.

```
vdem_df |>
  filter(year >= 2016, year <= 2022) |>
  group_by(year) |>
  summarise(n_gdp = sum(!is.na(gdp_pc)))
```

```
# A tibble: 7 x 2
   year n_gdp
  <dbl> <int>
1  2016   174
2  2017   174
3  2018   174
4  2019   174
5  2020     0
6  2021     0
7  2022     0
```

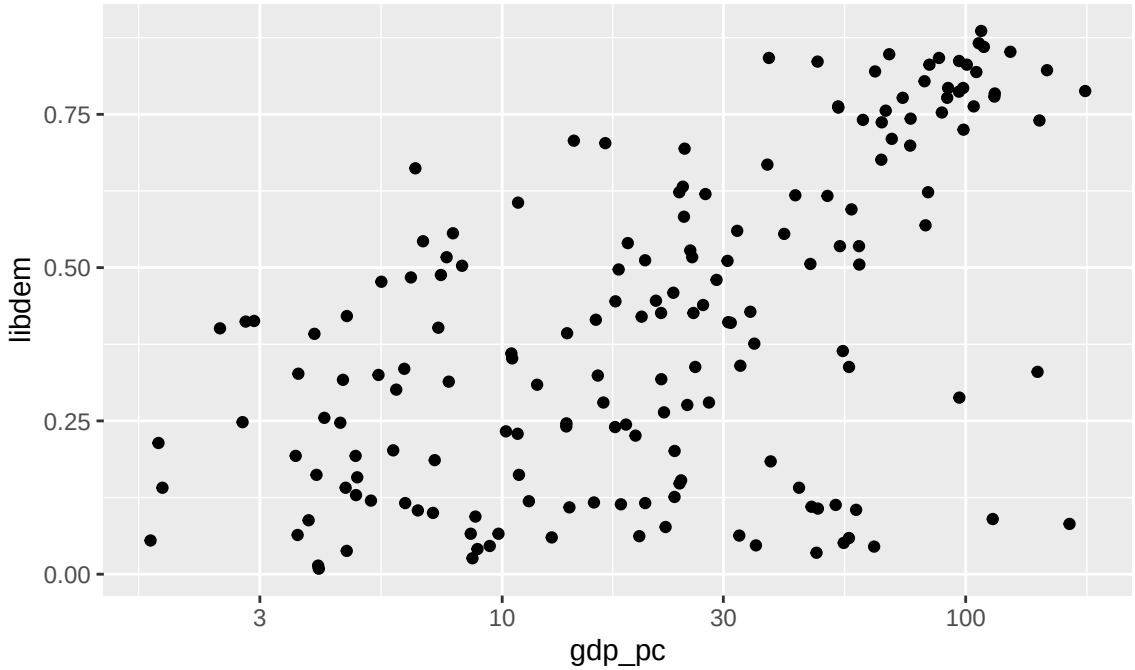
```
vdem_2019 <- vdem_df |>
  filter(year == 2019)
```

`gdp_pc` is known for 174 countries in every year up to 2019 and for none after. `sum(!is.na(x))` counts the non-missing values, because `TRUE` counts as 1.

2 The grammar of graphics

1. Build a scatter plot of `libdem` against `gdp_pc` in `vdem_2019` the way we did in class, one layer at a time: the data, then the aesthetics, then a geom, then `scale_x_log10()`. Run the code after every layer, and keep only the final figure in the chunk.

```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem)) +
  geom_point() +
  scale_x_log10()
```



2. In one sentence each: what did each of the four layers add?

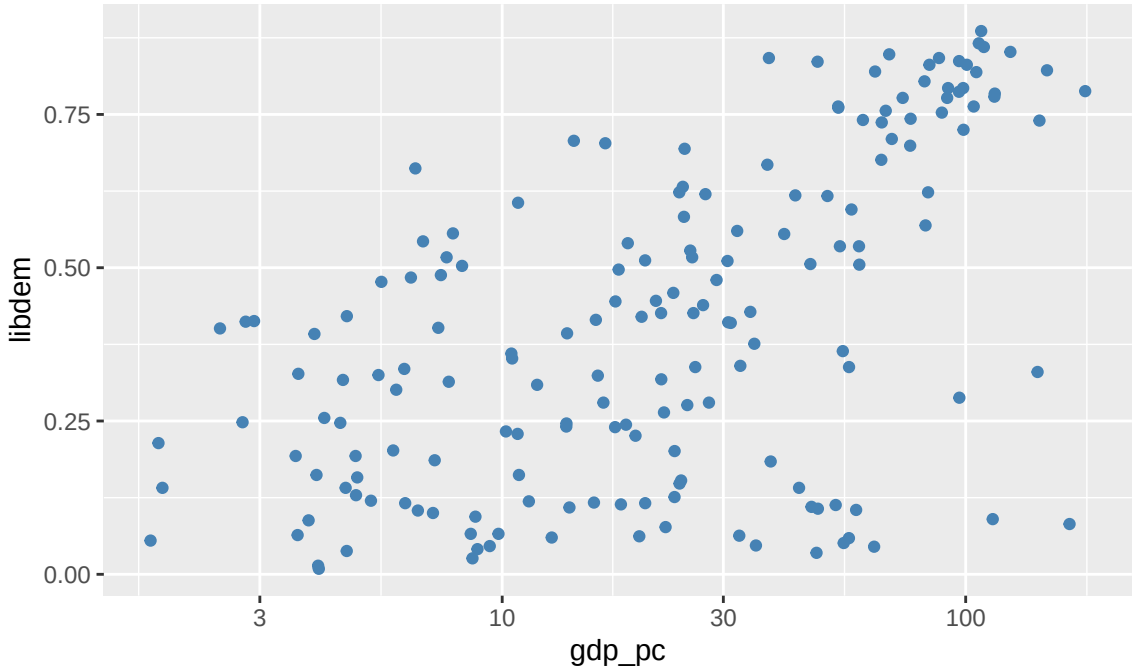
- **Data:** `ggplot()` with only the data draws an empty grey canvas, because nothing says yet what goes where.
- **Aesthetics:** `aes()` maps `gdp_pc` to the x-axis and `libdem` to the y-axis, so the axes and their ranges appear, but still no marks.
- **Geom:** `geom_point()` draws one point per country.
- **Scale:** `scale_x_log10()` puts the x-axis on a log scale, so the poorer countries are no longer squeezed together on the left.

3. Why does a log scale suit GDP per capita? Look at the figure with and without it.

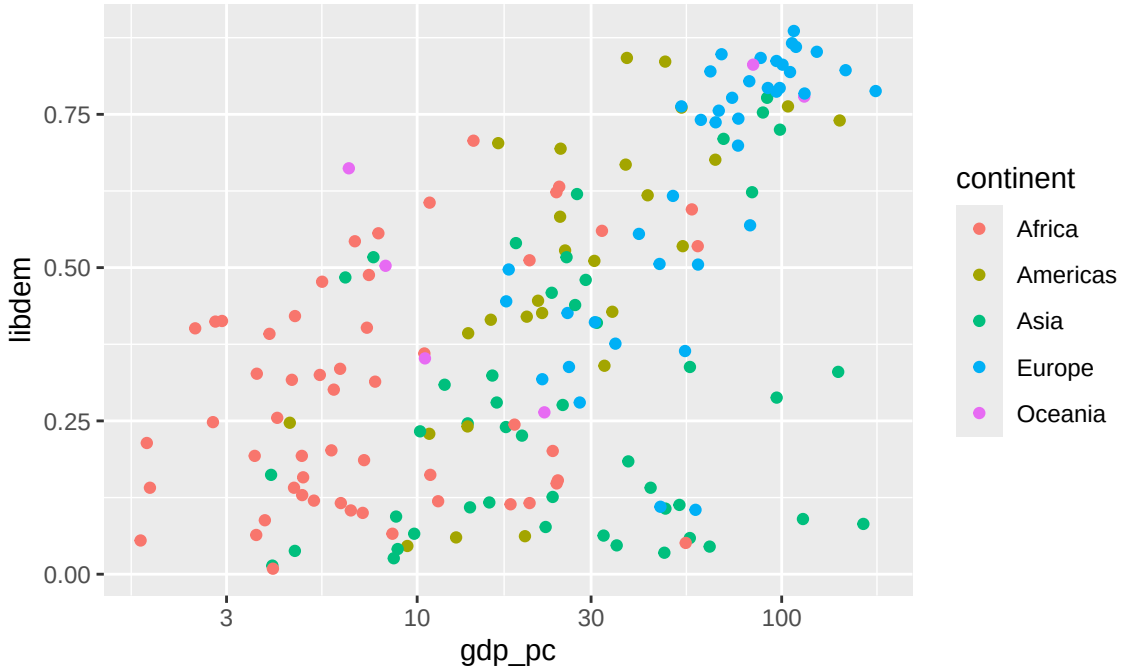
GDP per capita is strongly right-skewed: a few very rich countries such as Qatar stretch the axis, and most countries crowd into its left quarter. On a log scale, equal ratios get equal distances, so the step from 2 to 4 is as long as the step from 50 to 100. That is also the only comparison `gdp_pc` supports, because its unit is not dollars: "twice as rich" means something, and "20 units richer" does not.

4. Draw the same scatter plot twice: once with every point in one fixed colour of your choice, and once with each continent in its own colour. Which of the two sets `colour` inside `aes()`, and which outside?

```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem)) +
  geom_point(colour = "steelblue") +
  scale_x_log10()
```



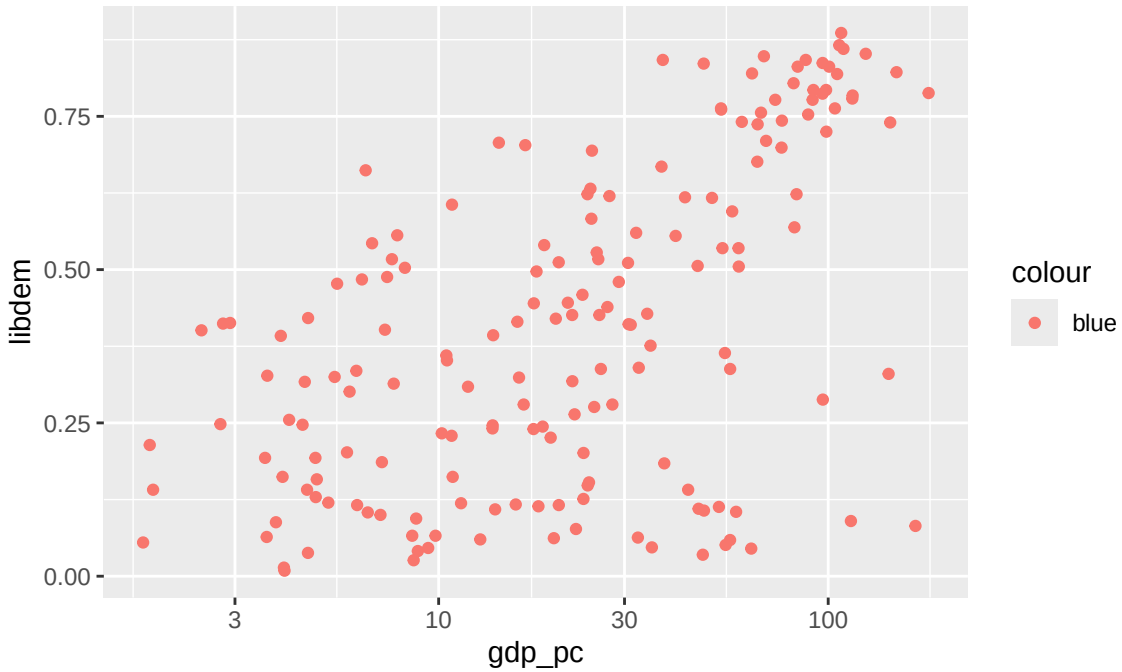

```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem, colour = continent)) +
  geom_point() +
  scale_x_log10()
```



A fixed colour for every point goes **outside** `aes()`, as an argument of the geom. A colour that depends on a variable goes **inside** `aes()`, and `ggplot2` adds a legend that explains the mapping.

5. **Before running anything**, predict what `aes(colour = "blue")` does. Then run it and explain what you see.

```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem, colour = "blue")) +
  geom_point() +
  scale_x_log10()
```

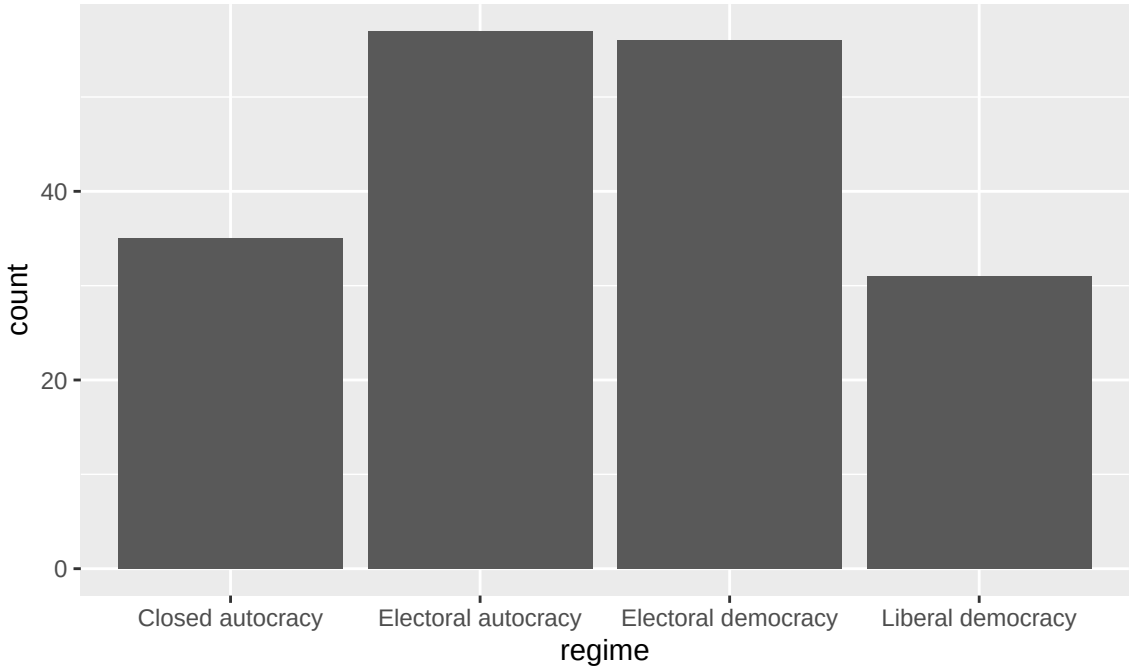


The points are not blue but salmon red, and a legend with the single entry "blue" appears. Inside `aes()`, `"blue"` is not a colour but a **value**: a constant variable that is `"blue"` for every country. `ggplot2` maps that one category to the first colour of its default palette, as it would map a continent.

3 One variable

1. Draw a bar chart of `regime` in `vdem_2025`. The bars follow the order of the factor levels set in the setup chunk, from the least to the most democratic regime. Here the alphabet happens to give the same order. Why is it still worth setting the order explicitly?

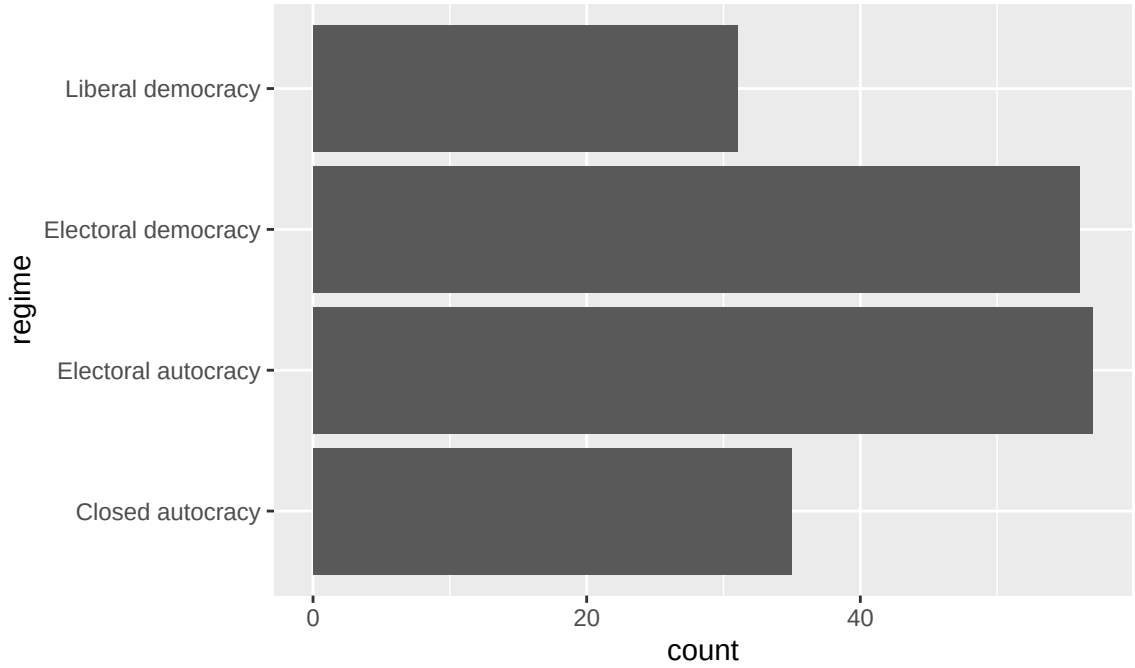
```
vdem_2025 |>
  ggplot(aes(x = regime)) +
  geom_bar()
```



`regime` is ordinal: its categories have a meaningful order, from closed autocracy to liberal democracy. The alphabet only matches that order by coincidence. Relabel one category, for example as "Autocracy (closed)" or in German, and the alphabetical order would scramble the bars without any warning. With factor levels, the order is a decision written into the data, and it survives any relabelling.

2. The category names overlap on the x-axis. Map `regime` to `y` instead of `x` and see what happens.

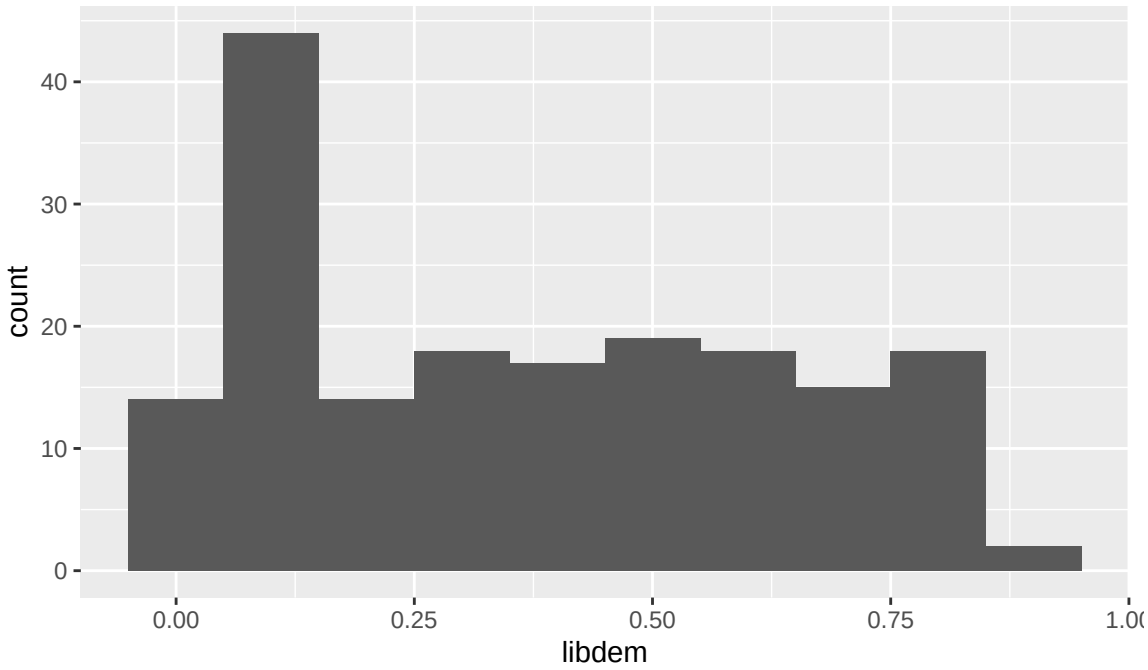
```
vdem_2025 |>
  ggplot(aes(y = regime)) +
  geom_bar()
```



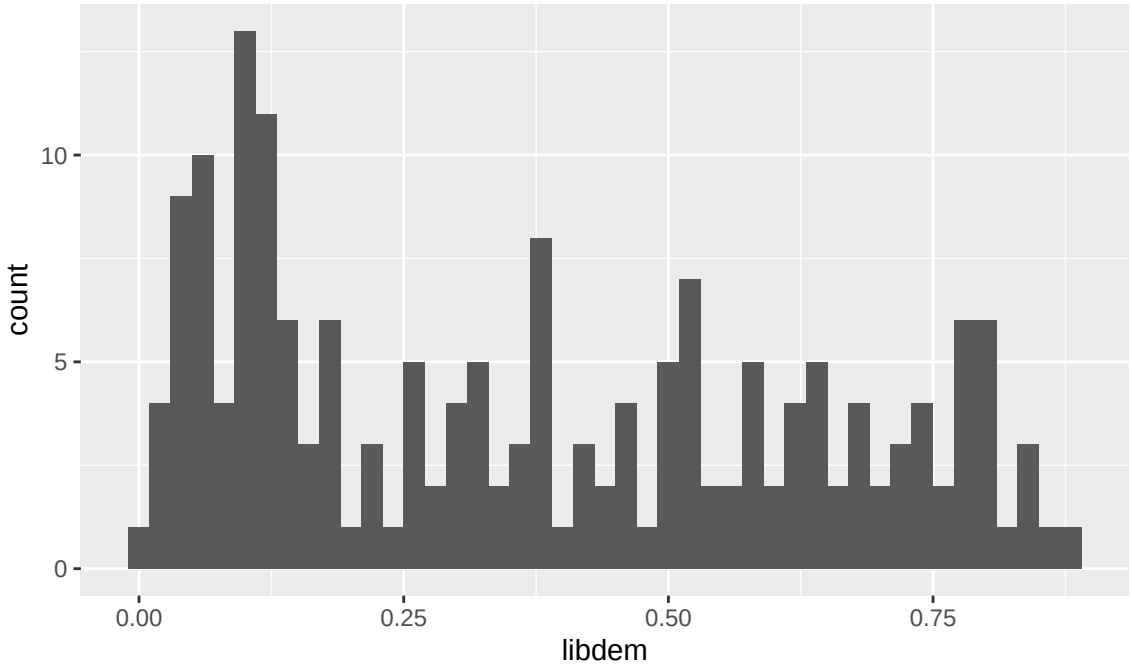
The bars turn horizontal and the labels have the whole width to themselves. The first level now sits at the bottom, so the most democratic regime is on top.

3. Draw a histogram of `libdem` in `vdem_2025`, once with `binwidth = 0.1` and once with `binwidth = 0.02`. Describe the shape of the distribution in one or two sentences. Which bin width shows it better, and why?

```
vdem_2025 |>
  ggplot(aes(x = libdem)) +
  geom_histogram(binwidth = 0.1)
```



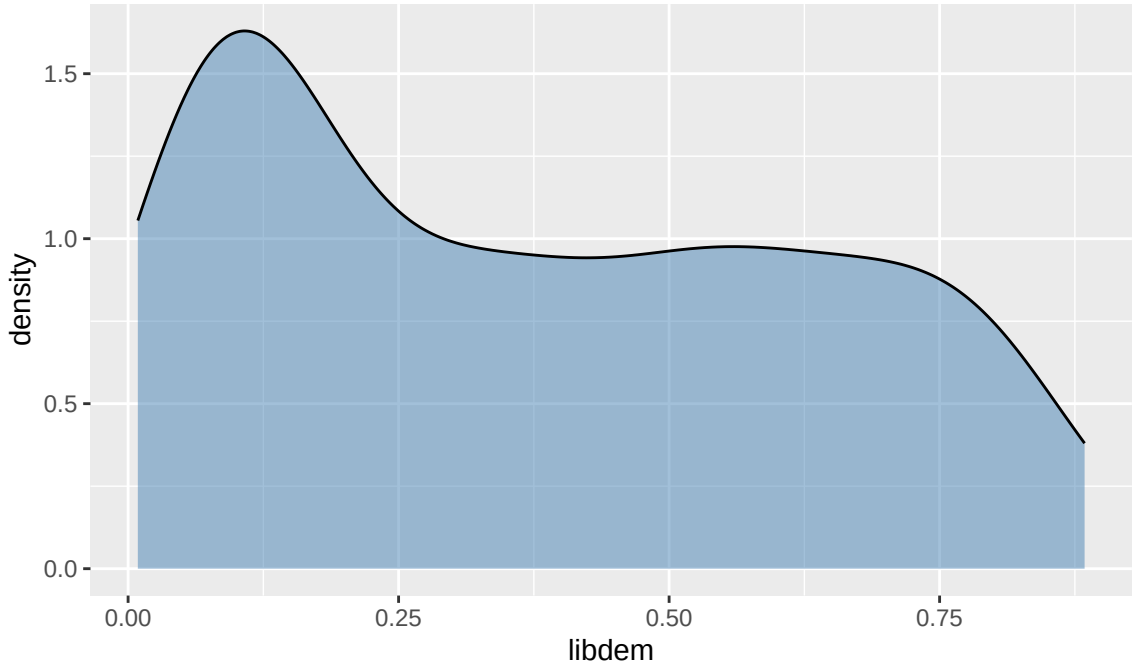

```
vdem_2025 |>
  ggplot(aes(x = libdem)) +
  geom_histogram(binwidth = 0.02)
```



The distribution is far from bell-shaped. Most countries pile up at the bottom, below 0.2, and the rest spread fairly evenly up to about 0.85, with no country above 0.9. The wider bins show this overall shape clearly. The narrow bins hold only a handful of countries each, so their heights are mostly noise, which is a lesson in itself: with 179 countries, a histogram cannot afford many bins.

4. Draw a density plot of `libdem` in `vdem_2025`, filled with a colour of your choice, with `alpha = 0.5`. What does the density plot show less clearly than the histogram?

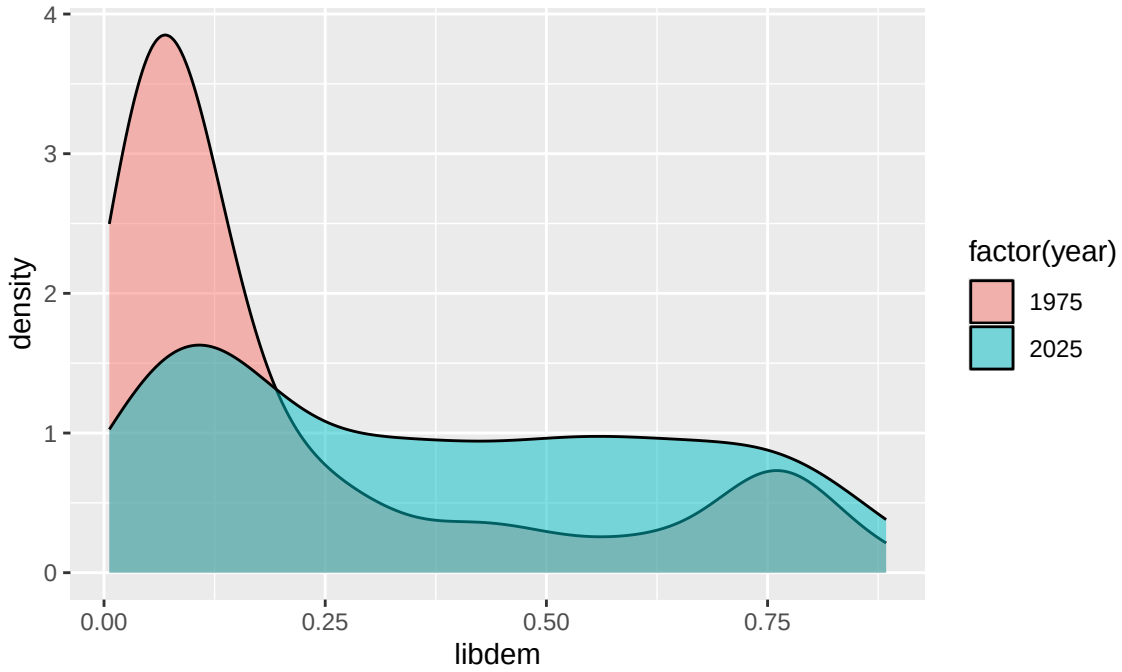
```
vdem_2025 |>
  ggplot(aes(x = libdem)) +
  geom_density(fill = "steelblue", alpha = 0.5)
```



The density plot hides **how many** countries there are: its y-axis is a density, not a count. It also smooths over the hard edges of the index, so the curve tails off below 0 and above the highest country, where no country can be.

5. **Stretch:** compare the distribution of `libdem` in 1975 with the one in 2025 in one density plot. Hint: filter both years, and map `fill = factor(year)`.

```
vdem_df |>
  filter(year %in% c(1975, 2025)) |>
  ggplot(aes(x = libdem, fill = factor(year))) +
  geom_density(alpha = 0.5)
```

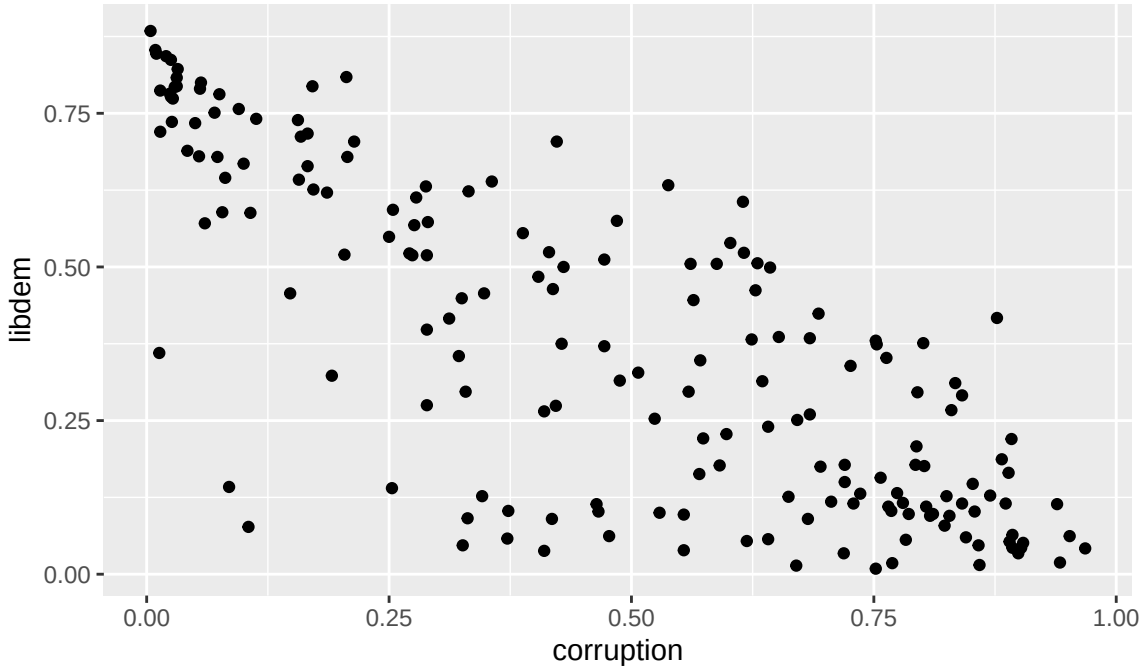


In 1975 almost all of the mass sits near zero, with a small bump of established democracies at the top. By 2025 the mass has spread to the middle and the upper range: the world is far more democratic than in 1975, even after the recent declines. `factor(year)` is needed because `year` is a number, and a number on `fill` gets a continuous colour scale instead of two distinct colours.

4 Two variables

1. Draw a scatter plot of `libdem` against `corruption` in `vdem_2025`. Describe the relationship in one or two sentences, and remember which way `corruption` runs.

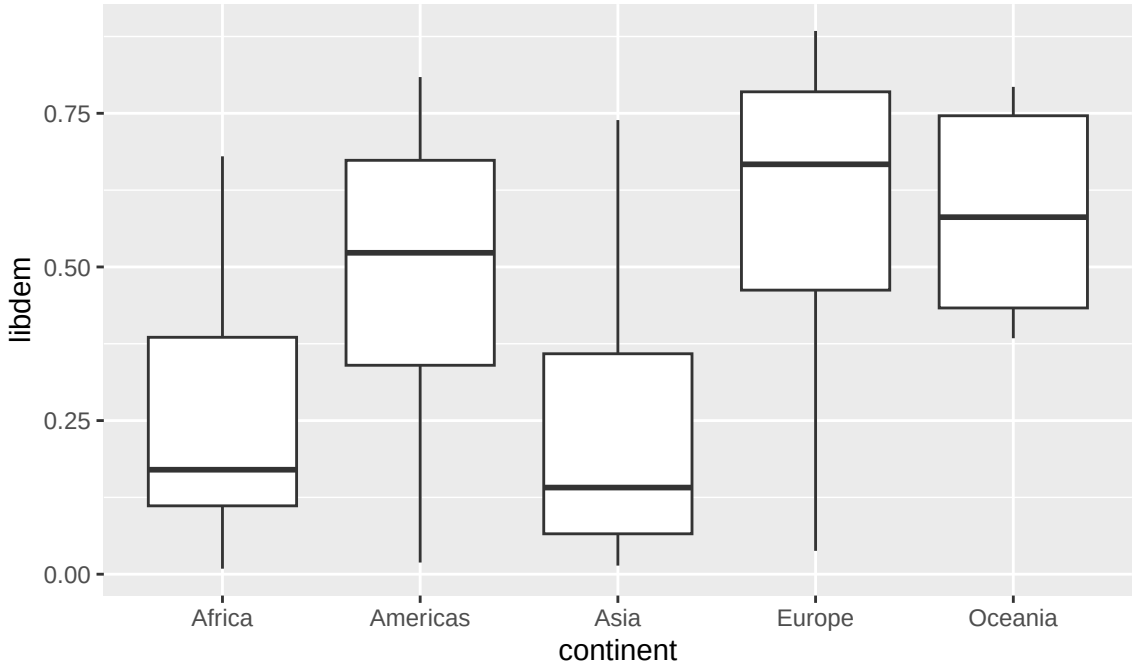
```
vdem_2025 |>
  ggplot(aes(x = corruption, y = libdem)) +
  geom_point()
```



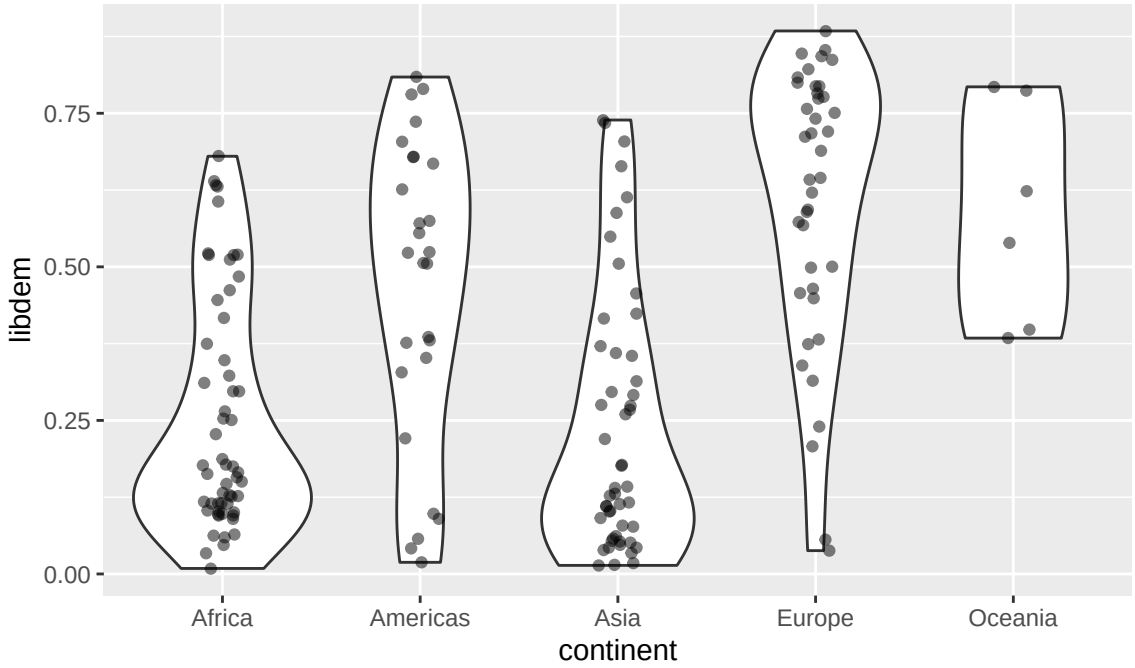
The relationship is strongly negative: the more corrupt a country's politics, the lower its liberal democracy score. Because `corruption` runs the other way from every other index, a negative slope means that liberal democracy and **clean** politics go together. The figure shows an association across countries, not that one causes the other.

2. Draw a box plot of `libdem` by `continent` in `vdem_2025`. Then draw the same data as a violin, with the countries as jittered points on top. What does the violin with points show that the box plot hides?

```
vdem_2025 |>
  ggplot(aes(x = continent, y = libdem)) +
  geom_boxplot()
```



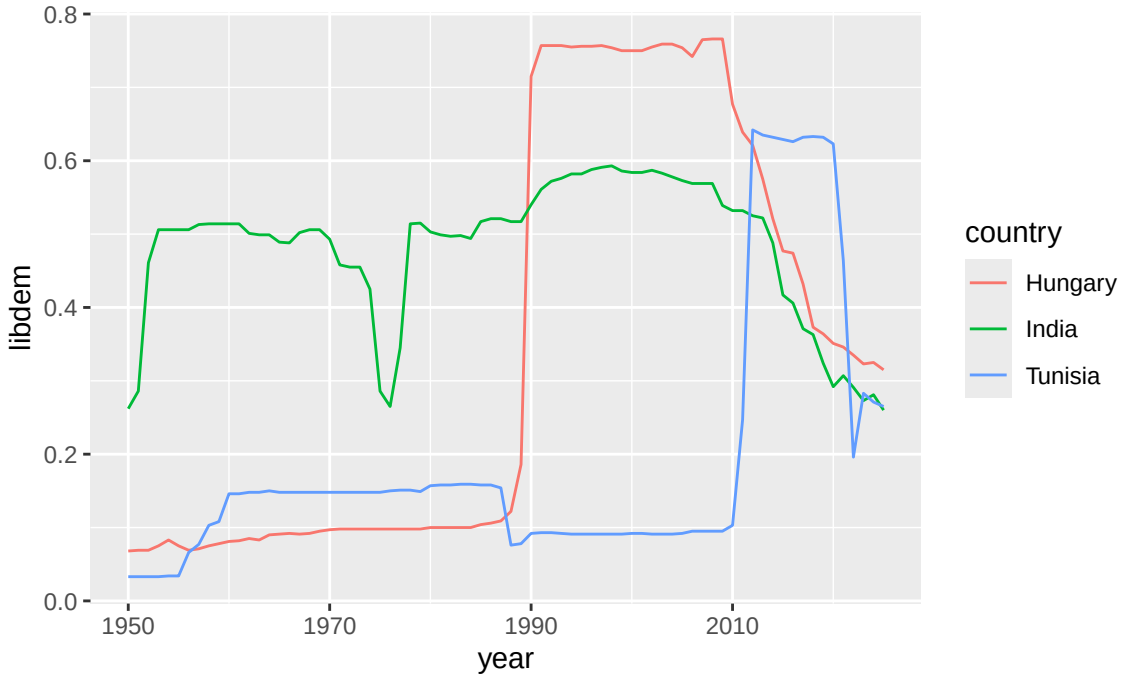

```
vdem_2025 |>
  ggplot(aes(x = continent, y = libdem)) +
  geom_violin() +
  geom_jitter(width = 0.1, alpha = 0.5)
```



The points show **how many** countries stand behind each box: Oceania has only 6, Africa has 56, and the box plot draws them the same way. The violin also shows the shape inside the box: in Africa and Asia most countries crowd at the bottom with a thin tail of democracies above, which a box with its whisker reduces to one line. `set.seed()` in the setup chunk keeps the jitter the same on every render.

3. Pick three countries that interest you and draw a line plot of their `libdem` over time, one colour per country. Hint: `%in%` and `colour = country`, as in class. Describe one turning point you see, and what you think happened.

```
vdem_df |>
  filter(country %in% c("Hungary", "India", "Tunisia")) |>
  ggplot(aes(x = year, y = libdem, colour = country)) +
  geom_line()
```

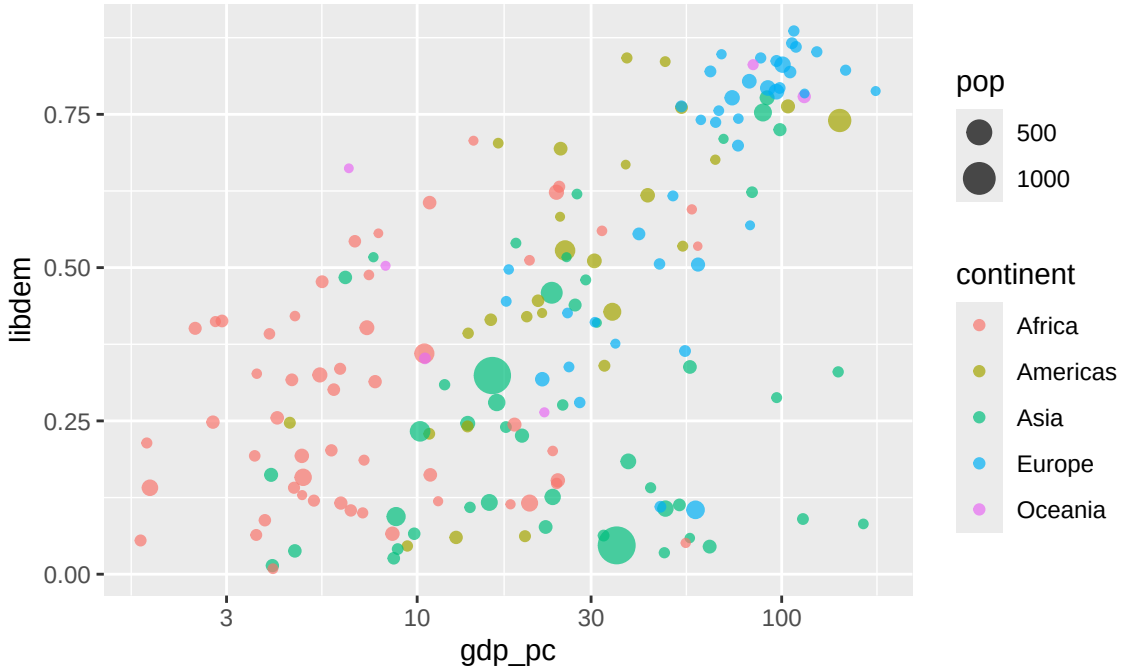


Tunisia has the sharpest turning points: its score jumps after the revolution of 2011, which started the Arab Spring, and collapses again after President Saied suspended parliament in 2021. Hungary democratised around 1990, at the end of communism, and has declined steadily since about 2010. India declines from the mid-2010s. Any three countries are a fine answer, as long as the turning point is read from the figure.

5 Three and more variables

1. Extend the scatter plot of `libdem` against `gdp_pc` in `vdem_2019`: map `continent` to colour and `pop` to point size. How many variables does the figure show now?

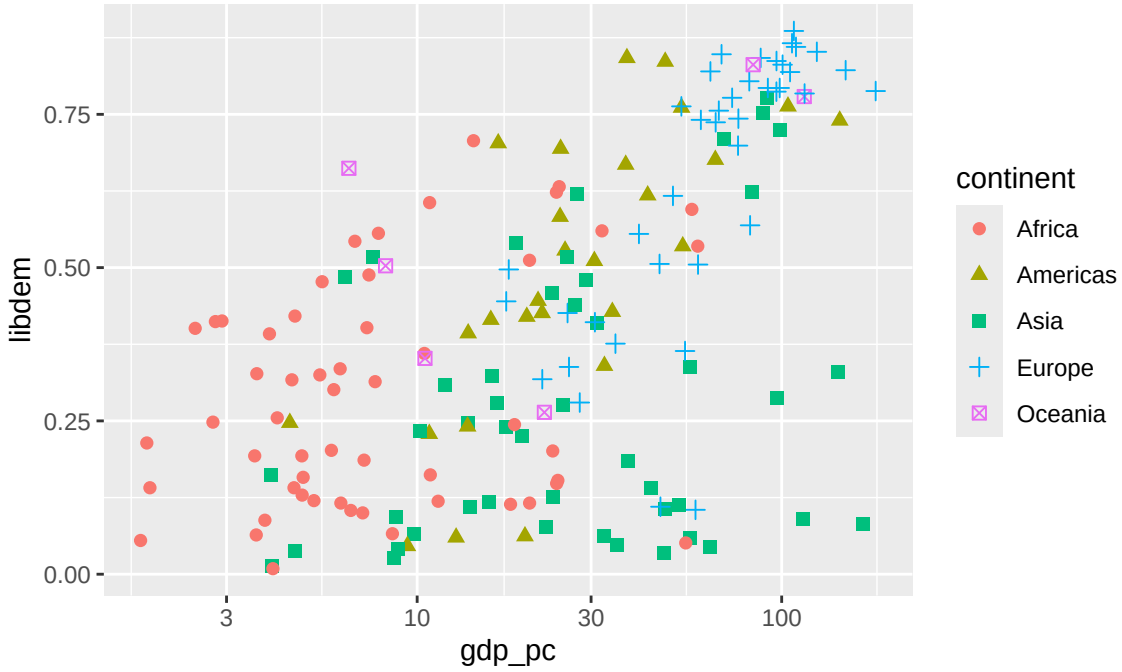
```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem, colour = continent, size = pop)) +
  geom_point(alpha = 0.7) +
  scale_x_log10()
```



Four: two positions (`gdp_pc` and `libdem`), a colour (`continent`) and a size (`pop`). The two very large points in the middle of the income range are China and India, and they sit far apart on the democracy axis.

2. Encode `continent` redundantly, with colour and point shape. Why does that help?

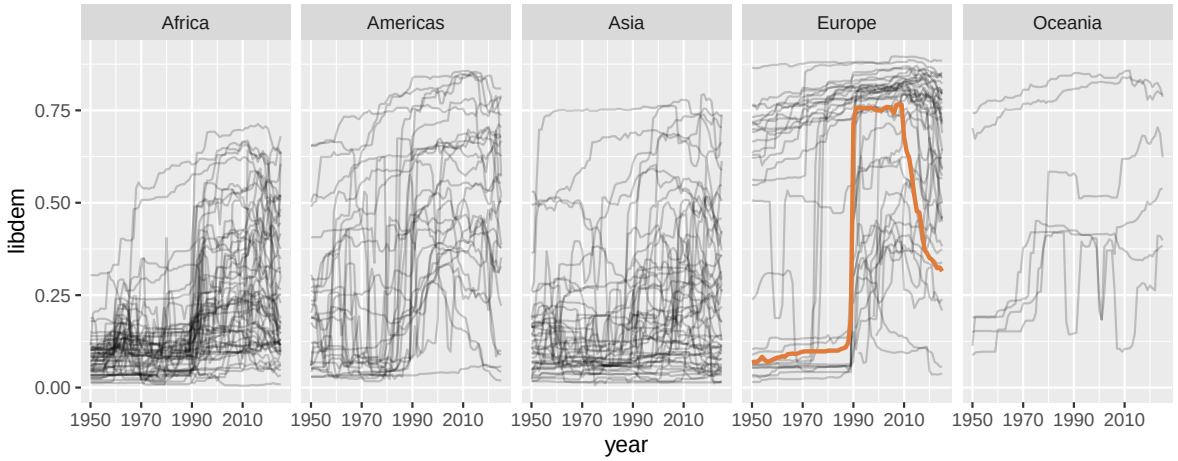
```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem, colour = continent, shape = continent)) +
  geom_point(size = 2) +
  scale_x_log10()
```



With two cues for one variable, a reader who cannot tell two colours apart can still tell the shapes apart, and so can anyone looking at a black and white print. Colour is never the only cue.

3. Plot `libdem` over time for **every** country, one line per country, with one panel per continent. Use `group = country` and a low `alpha`.
4. **Stretch:** draw one country of your choice on top, in a colour that stands out against the others. Hint: a second `geom_line()` with its own `data =`.

```
vdem_df |>
  ggplot(aes(x = year, y = libdem, group = country)) +
  geom_line(alpha = 0.2) +
  geom_line(
    data = filter(vdem_df, country == "Hungary"),
    colour = "#E07B39",
    linewidth = 1
  ) + # stretch
  facet_wrap(~continent, nrow = 1)
```



5. In two sentences: which continents saw the largest changes, and when?

Europe and the Americas changed most: Southern Europe and Latin America democratised in the 1970s and 1980s, and Eastern Europe around 1990, which is the jump in Hungary's line. Africa's lines lift around 1990 as well, and since about 2010 lines on most continents bend down again, which political scientists call a wave of autocratisation.

6 Styling: less ink, more information

1. Make the faceted line plot from Section 5 publication-ready:
    - a title that states a finding, and a subtitle that says what is plotted
    - axis labels with units, or no label where the axis speaks for itself
    - a caption with the source: `"Source: V-Dem dataset v16."`
    - `theme_classic()`
    - if you did the stretch, a label for your country with `geom_text()`

```
vdem_df |>
  ggplot(aes(x = year, y = libdem, group = country)) +
  geom_line(alpha = 0.2) +
  geom_line(
    data = filter(vdem_df, country == "Hungary"),
    colour = "#E07B39",
    linewidth = 1
  ) +
  geom_text(
    data = filter(vdem_df, country == "Hungary", year == 2000),
    aes(label = country),
```

```
    colour = "#E07B39",
    vjust = -1
  ) +
  facet_wrap(~continent, nrow = 1) +
  labs(
    title = "Democracy spread on every continent, and has receded since 2010",
    subtitle = "Liberal democracy index, one line per country, 1950 to 2025",
    x = NULL,
    y = "Liberal democracy (0 to 1)",
    caption = "Source: V-Dem dataset v16."
  ) +
  theme_classic()
```

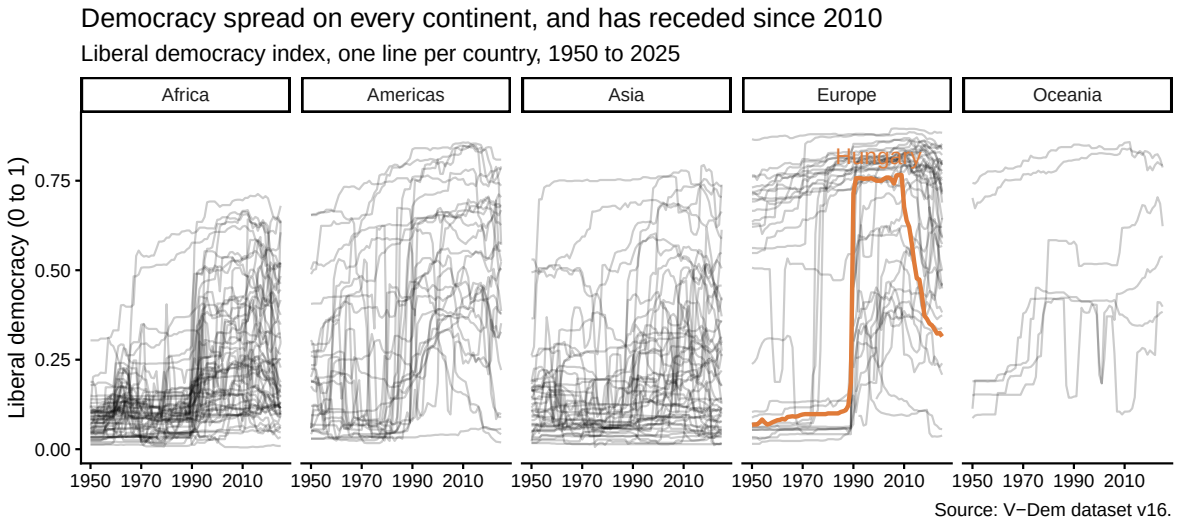


Figure 1: Liberal democracy index (0 to 1) for every country since 1950, one panel per continent, with Hungary in orange. Most countries are more democratic today than in 1950, and Hungary rose around 1990 and has declined since about 2010.

The year axis needs no label, because the numbers say what it is. The y-axis label states the scale, since "libdem" means nothing to a reader.

2. Annotate the scatter plot of `libdem` against `gdp_pc` in `vdem_2019`:

   - every country in one muted colour, such as a grey
   - a dashed horizontal line at the median of `libdem`, labelled with `annotate()`
   - one country of your choice highlighted in a contrasting colour, labelled with its name
   - `theme_classic()` and readable labels

```r
median_libdem <- median(vdem_2019$libdem, na.rm = TRUE)
qatar <- vdem_2019 |>
  filter(country == "Qatar")

vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem)) +
  geom_point(colour = "grey60") +
  geom_hline(yintercept = median_libdem, linetype = "dashed") +
  annotate("text", x = 1, y = median_libdem + 0.04,
           label = "Median", hjust = 0) +
  geom_point(data = qatar, colour = "#E07B39") +
  geom_text(data = qatar, aes(label = country),
            colour = "#E07B39", hjust = 1.2) +
  scale_x_log10() +
  labs(
    x = "GDP per capita (model-based estimate, log scale)",
    y = "Liberal democracy (0 to 1)"
  ) +
  theme_classic()
```

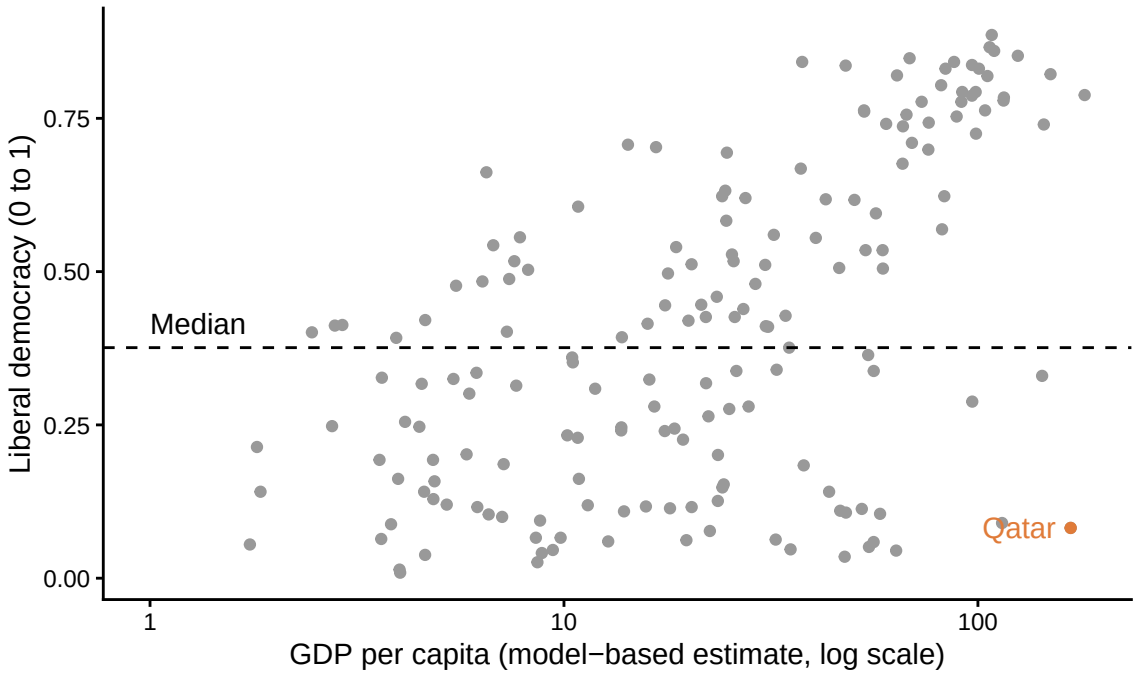
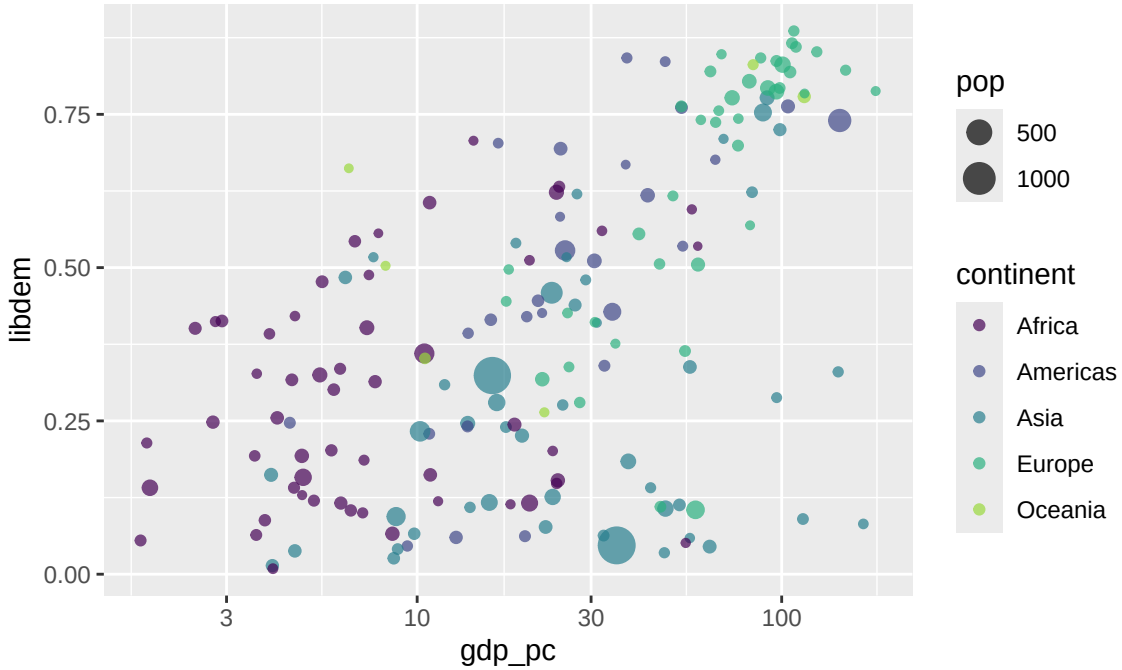


Figure 2: Liberal democracy against GDP per capita (log scale) in 2019, with the median democracy score dashed. Qatar, in orange, is among the richest countries and among the least democratic, so wealth alone does not make a democracy.

`median()` needs `na.rm = TRUE`, as in week 4, because some countries have no score.

3. Redo the scatter plot with colour and size from Section 5 with a colourblind-safe palette: `scale_colour_viridis_d(end = 0.85)`. Why `end = 0.85`? Try the plot without it.

```
vdem_2019 |>
  ggplot(aes(x = gdp_pc, y = libdem, colour = continent, size = pop)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  scale_colour_viridis_d(end = 0.85)
```



The viridis palette ends in a bright yellow that almost disappears on a white background. `end = 0.85` stops the palette before it reaches that yellow, so the last continent stays visible.

7 Common pitfalls: what is wrong here?

The two figures below are deliberately bad. Run them, and use the checklist from the end of the week 5 slides.

```
vdem_df |>
  group_by(continent, year) |>
  summarise(mean_libdem = mean(libdem, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = year, y = mean_libdem, colour = continent)) +
  geom_line(linewidth = 1.5) +
  scale_colour_manual(
    values = c("red3", "green4", "darkorange3", "forestgreen", "firebrick")
  ) +
  theme_dark()
```

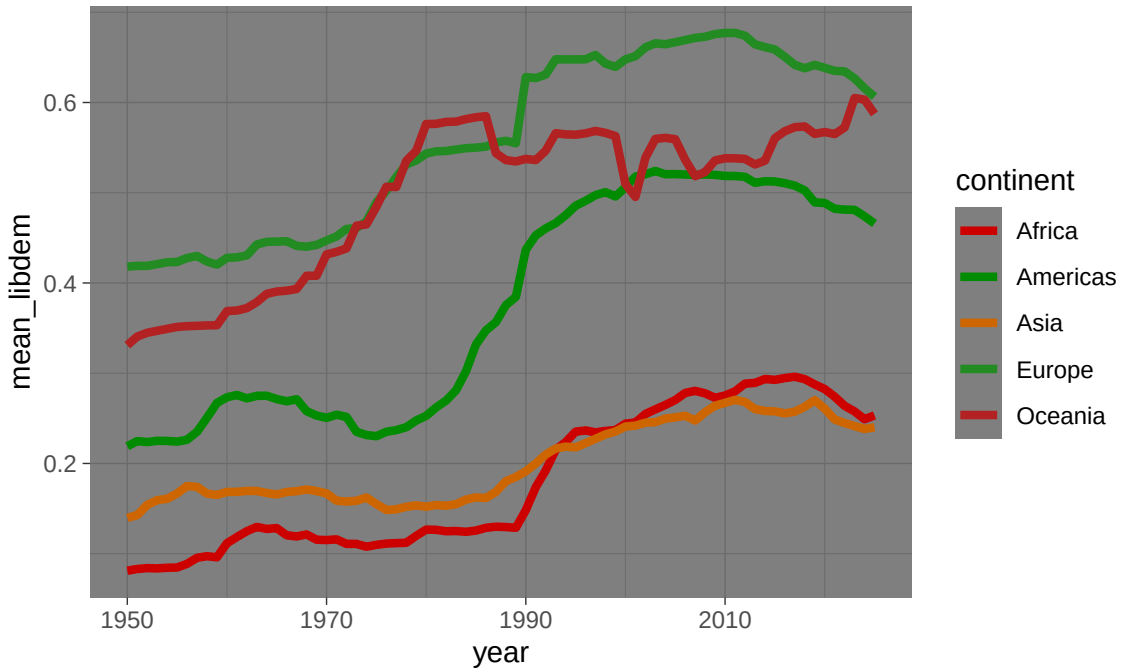


Figure 3: A deliberately bad figure: mean liberal democracy by continent since 1950.

```
vdem_df |>
  filter(year == 2025) |>
  group_by(continent) |>
  summarise(mean_libdem = mean(libdem, na.rm = TRUE)) |>
  ggplot(aes(x = continent, y = mean_libdem, fill = continent)) +
  geom_col() +
  coord_cartesian(ylim = c(0.2, 0.6))
```

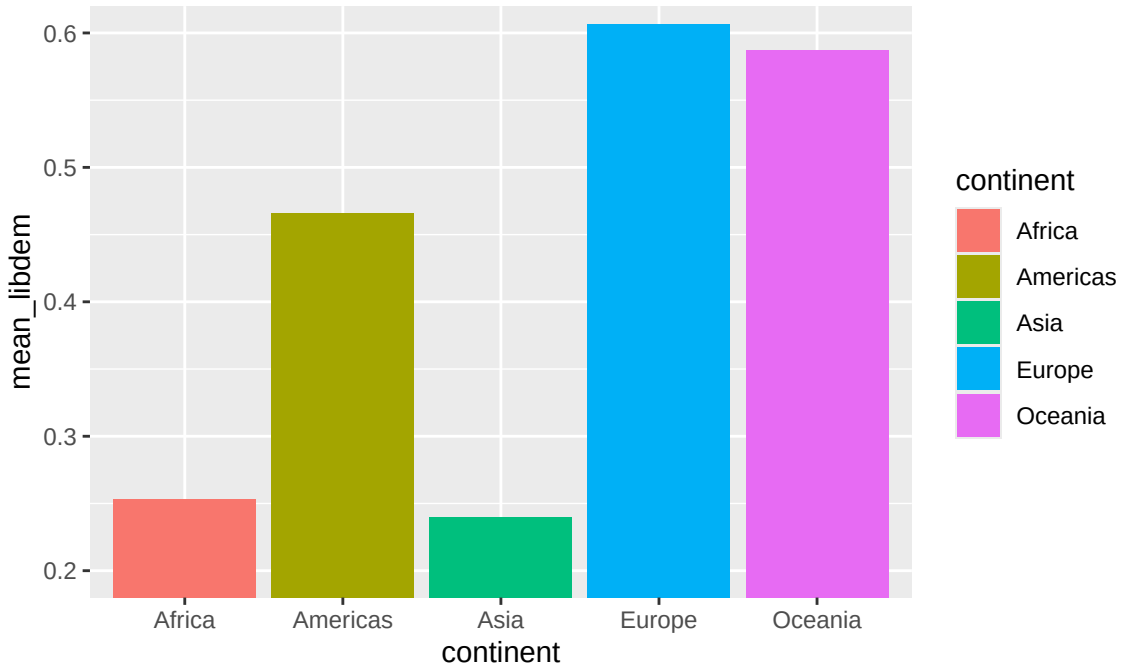


Figure 4: A deliberately bad figure: mean liberal democracy by continent in 2025.

1. List at least two problems in Figure 3 and at least two in Figure 4. For each, say what a reader would get wrong.

Figure 3:

- **Two reds and two greens.** A reader with a red-green colour vision deficiency cannot tell Africa, Asia, Europe and Oceania apart, and nobody can match the two greens to the legend quickly.
- **A dark theme.** The grey background lowers the contrast of every line and adds ink that carries no data.
- **No title and raw variable names.** A reader sees `mean_libdem` and `year` but not what the index is, on which scale, or what the figure is meant to show.
- **Thick lines** hide where the lines cross.

Figure 4:

- **A truncated axis.** The bars start at 0.2, not at 0, so their lengths are wrong. Europe's bar looks many times as long as Asia's, while its mean is only about two and a half times as large, and Africa's and Asia's bars nearly vanish. A bar encodes a value by its length, so a bar axis must start at zero.
- **A redundant legend and fill.** The colour repeats the x-axis labels and carries no information.

- **Alphabetical order.** The bars are sorted by name, not by value, so the ranking has to be read off bar by bar.
- **Raw labels** again: `mean_libdem` and `continent`.

2. Fix both figures. The cells below already carry figure labels. Replace each placeholder `fig-cap` with a caption that a reader can understand without the text.

```
continent_means <- vdem_df |>
  group_by(continent, year) |>
  summarise(mean_libdem = mean(libdem, na.rm = TRUE), .groups = "drop")

continent_means |>
  ggplot(aes(x = year, y = mean_libdem, colour = continent)) +
  geom_line(linewidth = 0.8) +
  geom_text(
    data = filter(continent_means, year == max(year)),
    aes(label = continent),
    hjust = -0.1,
    size = 3
  ) +
  scale_colour_viridis_d(end = 0.85, guide = "none") +
  scale_x_continuous(limits = c(1950, 2040)) +
  labs(
    title = "Democracy rose everywhere, then slipped on most continents",
    x = NULL,
    y = "Mean liberal democracy (0 to 1)",
    caption = "Source: V-Dem dataset v16."
  ) +
  theme_classic()
```

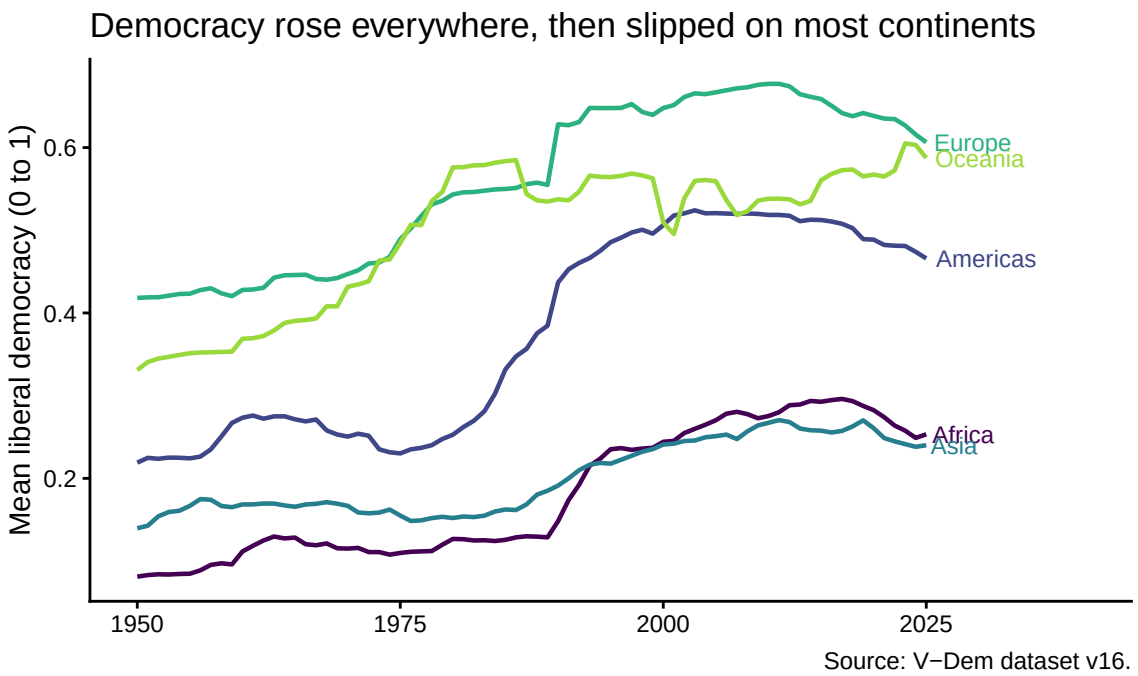


Figure 5: Mean liberal democracy index (0 to 1) of the countries on each continent, 1950 to 2025. Every continent is more democratic than in 1950, and every continent except Oceania has lost ground since about 2010.

A colourblind-safe palette, a white background, a finding as the title, and each line labelled at its end instead of through a legend, as on the slide about colours that hide the data. The x-axis runs to 2040 only to make room for the labels.

```
vdem_2025 |>
  group_by(continent) |>
  summarise(mean_libdem = mean(libdem, na.rm = TRUE)) |>
  ggplot(aes(x = mean_libdem, y = fct_reorder(continent, mean_libdem))) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Europe and Oceania are the most democratic continents",
    x = "Mean liberal democracy in 2025 (0 to 1)",
    y = NULL,
    caption = "Source: V-Dem dataset v16."
  ) +
  theme_classic()
```

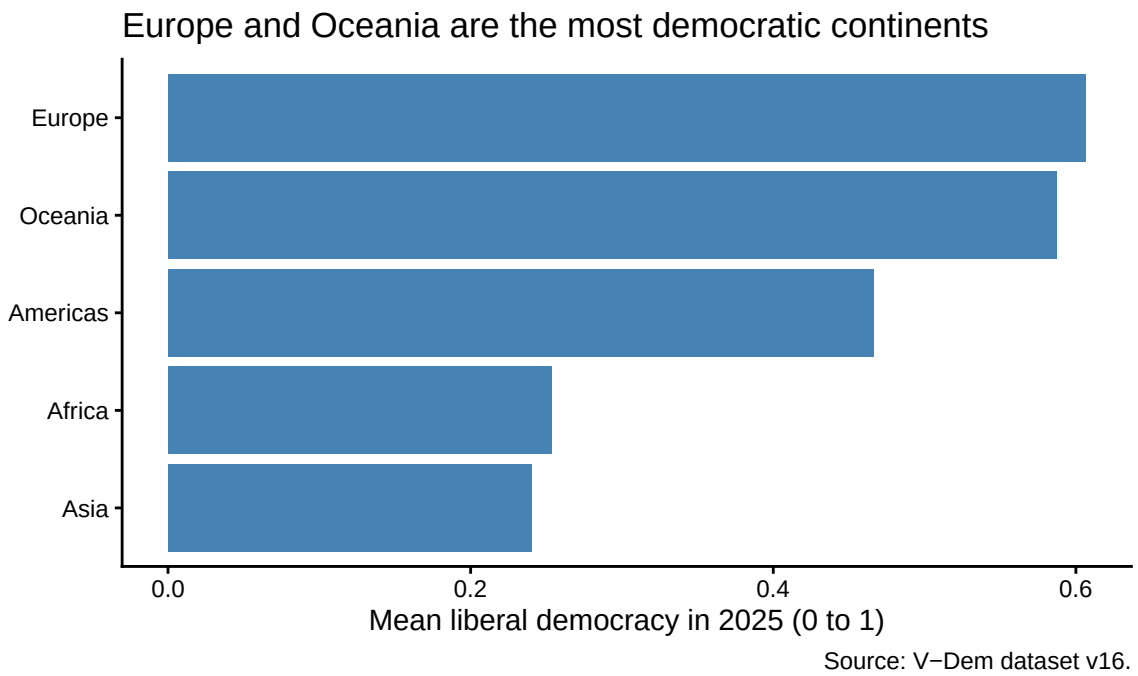


Figure 6: Mean liberal democracy index (0 to 1) of the countries on each continent in 2025, sorted from the most to the least democratic. Europe and Oceania score more than twice as high as Africa and Asia.

The axis starts at zero again, the bars are sorted by value with `fct_reorder()`, one colour replaces the redundant fill, and the bars are horizontal so the names read easily.

8 Your figure: a data summary

> **!** Important
>
> **This section is step 2 of the workflow, the data summary.** In week 4 the data summary was a table. Here it is a figure that answers a precise question, and a reader should understand it without your code.

The question. How has the share of countries in each regime type changed since 1950, and does the pattern differ by continent?

1. Write **one pipeline**, starting from `vdem_df`, that:
    - drops the rows without a `regime`

- counts the countries per `continent`, `year` and `regime`
- turns each count into a share of the countries on that continent in that year

Assign the result to `regime_shares`, and run the pipeline after every new line, as in class. Hint: `count(continent, year, regime)`, then `group_by(continent, year)` and `mutate(share = n / sum(n))`. End with `complete(continent, year, regime, fill = list(n = 0, share = 0))`: a regime type that no country had in some year has no row, and `geom_area()` needs one with a share of zero.

```
regime_shares <- vdem_df |>
  filter(!is.na(regime)) |>
  count(continent, year, regime) |>
  group_by(continent, year) |>
  mutate(share = n / sum(n)) |>
  ungroup() |>
  complete(continent, year, regime, fill = list(n = 0, share = 0))

regime_shares
```

```
# A tibble: 1,520 x 5
   continent  year regime                 n  share
   <chr>     <dbl> <fct>              <int>  <dbl>
 1 Africa     1950 Closed autocracy      49 0.925
 2 Africa     1950 Electoral autocracy    4 0.0755
 3 Africa     1950 Electoral democracy    0 0
 4 Africa     1950 Liberal democracy      0 0
 5 Africa     1951 Closed autocracy      50 0.926
 6 Africa     1951 Electoral autocracy    4 0.0741
 7 Africa     1951 Electoral democracy    0 0
 8 Africa     1951 Liberal democracy      0 0
 9 Africa     1952 Closed autocracy      50 0.926
10 Africa     1952 Electoral autocracy    4 0.0741
# i 1,510 more rows
```

`mutate()` after `group_by()` computes `sum(n)` within each continent and year, so the shares add up to 1 in every group. `ungroup()` keeps the grouping from surprising a later step. Without `complete()`, a regime type disappears from a continent for a few years, `geom_area()` draws straight across the gap, and the stack overshoots 100 percent.

2. Plot `regime_shares` as a figure for a reader: the share on the y-axis, the year on the x-axis, `regime` as the fill colour, and one panel per continent. `geom_area()` stacks the shares on top of each other, as a stacked bar chart stacks counts. Style it as in Section 6,

with a viridis palette. The cell below already carries the label `fig-regimes`. Replace its placeholder `fig-cap` with a caption that a reader can understand without the text.

```
regime_shares |>
  ggplot(aes(x = year, y = share, fill = regime)) +
  geom_area() +
  facet_wrap(~continent, nrow = 1) +
  scale_fill_viridis_d(end = 0.95) +
  scale_y_continuous(labels = scales::label_percent()) +
  scale_x_continuous(breaks = c(1960, 1990, 2020)) +
  labs(
    title = "Closed autocracy retreated, but liberal democracy did not take its place",
    x = NULL,
    y = "Share of countries",
    fill = NULL,
    caption = "Source: V-Dem dataset v16, Regimes of the World."
  ) +
  theme_classic() +
  theme(legend.position = "bottom")
```

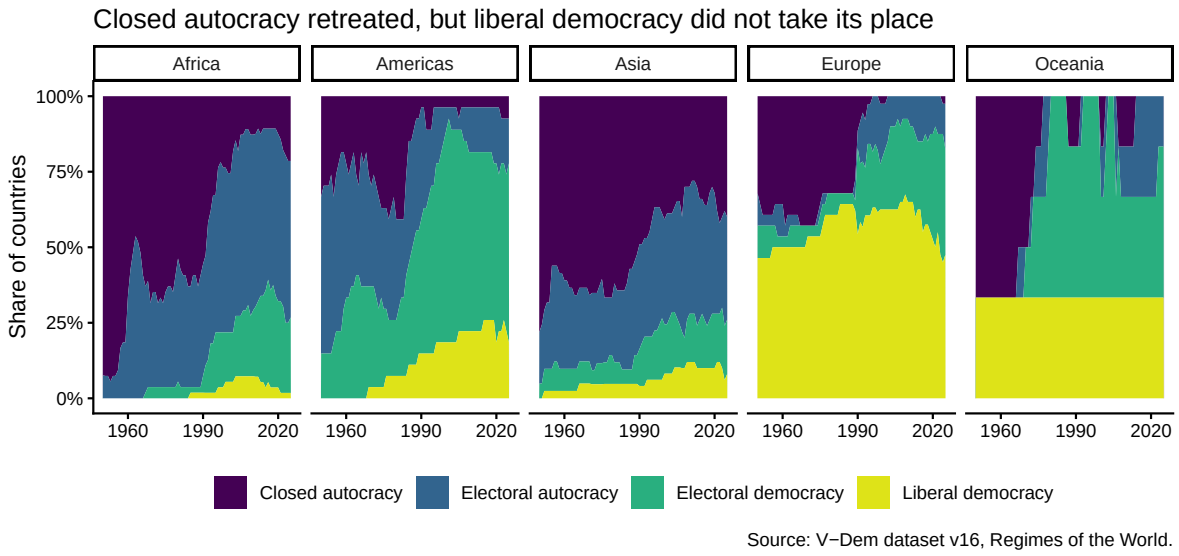


Figure 7: Share of countries in each regime type by continent, 1950 to 2025, with each country counted once. Closed autocracies, dark at the top, shrank on every continent and gave way mostly to electoral regimes, while liberal democracies, yellow at the bottom, remain a minority outside Europe and Oceania.

3. Below the figure, write three to four sentences that answer the question.

- Reference the figure with `@fig-regimes`.
- Put at least one number into a sentence with inline code, so that it updates when the data does.
- Name one thing the figure hides, or one reason to read it with care. Think about what one country counts for (India and Fiji count the same), and about which units V-Dem counts in 1950, when much of Africa and Asia was still under colonial rule.

Figure 7 shows that closed autocracy, the regime without multiparty elections, has retreated on every continent: in Africa its share fell from 92% of the units V-Dem codes in 1950 to 21% in 2025. It was replaced mostly by electoral autocracy and electoral democracy, not by liberal democracy, which holds 2% of African and 8% of Asian countries in 2025, against 48% in Europe. The figure has to be read with care: every country counts once, so India weighs as much as Fiji, and Oceania's six countries make its shares jump in steps of a sixth. In 1950 many African and Asian units were still colonies, without elections for a government of their own, so part of the early change is decolonisation rather than democratisation.

> Tip
>
> **Why a stacked area and not five lines per panel?** The shares add up to 100 percent, and a stacked area shows that constraint directly: when one regime type grows, the reader sees which one shrank. Its weakness is that only the bottom band, liberal democracy here, starts from a common baseline, so the middle bands are harder to compare over time.

Reflection

Which figure took you longest, and why? Is there a part of `ggplot2` you are still unsure about?

The reflection is personal and has no sample answer. A good one names a concrete figure, such as the stacked areas that needed `group_by()` before `mutate()`, and says what made it hard.

Session info and execution time

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: Europe/Berlin
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] vdemdata_16.0   janitor_2.2.1   lubridate_1.9.5 forcats_1.0.1
 [5] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0
 [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.6.1     tidyselect_1.2.1
 [5] snakecase_0.11.1   scales_1.4.0       yaml_2.3.12        fastmap_1.2.0
 [9] R6_2.6.1           labeling_0.4.3     generics_0.1.4     knitr_1.52
[13] pillar_1.11.1      RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.3.0
[17] utf8_1.2.6         stringi_1.8.9      xfun_0.60          S7_0.2.2
[21] otel_0.2.0         viridisLite_0.4.3  timechange_0.4.0   cli_3.6.6
[25] withr_3.0.3        magrittr_2.0.5     digest_0.6.39      grid_4.6.1
[29] hms_1.1.4          lifecycle_1.0.5    vctrs_0.7.3        evaluate_1.0.5
[33] glue_1.8.1         farver_2.1.2       rmarkdown_2.32     tools_4.6.1
[37] pkgconfig_2.0.3    htmltools_0.5.9
```

```r
end_time <- Sys.time()
cat(
  "Total execution time:",
  round(difftime(end_time, start_time, units = "mins"), 1),
  "minutes\n"
)
```

```
Total execution time: 0.2 minutes
```